Permanent Mathematical Notes with a Deliberately Long Explanatory Title

Stable references for notes that continue to grow.

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Suggested citation

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Contents

Chapter 1

The first tagged chapter (Tag `0AB`)

1.1 A published section (Tag `0ABCD`)

1.1.1 Published statements (Tag `0ABCDS1`)

Theorem 1.1.1 (Fundamental comparison) (Tag `0ABCDEF`). If two objects satisfy the same universal property, they are uniquely isomorphic in the appropriate sense.

Definition 1.1.2 (Tag `0ABCDE1`). A published object is an entry whose semantic content and permanent tag have been recorded in the global registry.

Theorem 1.1.3 (Tag `0ABCDE2`). A permanent tag does not require the statement to have a title.

Published local references stay compact: Chapter 1 (Tag `0AB`), Section 1.1 (Tag `0ABCD`), Subsection 1.1.1 (Tag `0ABCDS1`), Theorem 1.1.1 (Tag `0ABCDEF`), and Definition 1.1.2 (Tag `0ABCDE1`). The old identifier remains queryable as the frozen record [Tag System Test Book, Theorem 1.1.0, retired Tag `0ABCDE0`; replaced by Tag `0ABCDEF`].

1.1.2 Mathematical alphabets (Tag `0ABCDS2`)

The inline specimen $\mathcal{M} \in \mathscr{C}$, $f : X \to Y$, $H^i(X, \mathcal{M}) \cong \mathbb{Q}^{\oplus r_i}$, and $\mathfrak{g} = \mathfrak{sl}_n(k)$ shows the selected families in running text. The full alphabet comparison is

$$\begin{aligned}
\text{Roman} &: \mathrm{ABCDEFGHIJKLMNOPQRSTUVWXYZ}, \\
& \mathrm{abcdefghijklmnopqrstuvwxyz}, \\
\textit{Italic} &: ABCDEFGHIJKLMNOPQRSTUVWXYZ, \\
& abcdefghijklmnopqrstuvwxyz, \\
\textbf{Bold} &: \mathbf{ABCDEFGHIJKLMNOPQRSTUVWXYZ}, \\
& \mathbf{abcdefghijklmnopqrstuvwxyz}, \\
\textsf{Sans} &: \mathsf{ABCDEFGHIJKLMNOPQRSTUVWXYZ}, \\
& \mathsf{abcdefghijklmnopqrstuvwxyz}, \\
\texttt{Mono} &: \mathtt{ABCDEFGHIJKLMNOPQRSTUVWXYZ}, \\
& \mathtt{abcdefghijklmnopqrstuvwxyz}.
\end{aligned}$$

The decorative and semantic families are

$$\begin{aligned}
\text{Calligraphic} &: \mathcal{ABCDEFGHIJKLMNOPQRSTUVWXYZ}, \\
& \mathcal{abcdefghijklmnopqrstuvwxyz}, \\
\text{Script} &: \mathscr{ABCDEFGHIJKLMNOPQRSTUVWXYZ}, \\
& \mathscr{abcdefghijklmnopqrstuvwxyz}, \\
\text{Blackboard} &: \mathbb{ABCDEFGHIJKLMNOPQRSTUVWXYZ}, \\
& \mathbb{abcdefghijklmnopqrstuvwxyz}, \\
\text{Fraktur} &: \mathfrak{ABCDEFGHIJKLMNOPQRSTUVWXYZ}, \\
& \mathfrak{abcdefghijklmnopqrstuvwxyz}.
\end{aligned}$$

Chapter 2

A draft chapter (Draft tag `0CD`)

2.1 Every theorem-style environment (Draft tag `0CDEF`)

2.1.1 Statements, proofs, steps, and cases (Draft tag `0CDEFS1`)

Slogan *Stable identity belongs to the mathematics, not to its current page number.*

Theorem A (The gallery theorem). Every supported statement style remains readable in long-form notes.

Proposition 2.1.1 (A moved statement) (Tag `0ABEF01`). A mathematically unchanged published entry keeps its tag after it moves.

Theorem 2.1.2. This draft theorem has no provisional tag, so its reference contains only the current theorem number.

Lemma 2.1.3 (Local cancellation). If $g \circ f$ and g are isomorphisms, then f is an isomorphism.

Corollary 2.1.4. An identity morphism is an isomorphism.

Conjecture 2.1.5 (A plausible extension). The same comparison principle persists in the derived setting.

Question 2.1.6. Which hypotheses can be weakened without changing the conclusion?

Example 2.1.7 (Draft tag `0CDEF01`). This valid-looking draft tag is visible but is not globally reserved.

Exercise 2.1.8 (Check the universal property). Prove the uniqueness assertion in Theorem 1.1.1 (Tag `0ABCDEF`).

Construction 2.1.9 (A fiber product). For maps $X \to S \leftarrow Y$, form $X \times_S Y$ by its universal property.

Notation 2.1.10. Write $D^b(X)$ for the bounded derived category of coherent sheaves on X.

Remark 2.1.11 (Draft tag `0CDEF01`). Duplicate provisional tags are permitted in a draft scope and must be resolved before publication.

Proof sketch for Theorem A. The proof can be divided into referenceable steps and cases.

Step 1 (Set up the comparison). Choose the canonical maps supplied by the two universal properties.

Claim 2.1.12 (The composites are identities). Both composites agree with the corresponding identity morphisms.

Case 1 (The source is initial). Uniqueness of a map from the source proves the first identity.

Case 2 (The target is terminal). Uniqueness of a map to the target proves the second identity.

Thus Step 1, Claim 2.1.12, and Case 1 complete the argument. □

Draft references omit their provisional tags: Chapter 2, Section 2.1, Subsection 2.1.1, Theorem 2.1.2, Example 2.1.7, and Remark 2.1.11.

2.2 Typesetting features (Draft tag `0CDGH`)

An aligned equation can carry an ordinary label:

$$Lg^* Rf_* \mathcal{F} \simeq Rf'_* Lg'^* \mathcal{F}. \tag{2.1}$$

The reference eq. (2.1) uses the standard cleveref behavior.

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "g"'] & Y \end{tikzcd}$$

Nested lists use alphabetic and roman labels:

(a) Verify existence.

(b) Verify uniqueness.

  (i) compare the two candidates;

  (ii) apply the universal property.

Scope	Tag length	Example
Book	1	`0`
Chapter	3	`0AB`
Section	5	`0ABCD`
Entry or subsection	7	`0ABCDEF`

```python
def compose(f, g):
    """Return the composite g after f."""
    return lambda x: g(f(x))
```

Listing 2.1: A small source-code listing

Both Code 2.1 and citations such as [Har77] are available without loading extra packages in the note source.

Bibliography

[Har77] Robin Hartshorne. *Algebraic Geometry*. Springer, 1977 (cit. on p. 8).